

EXPERIMENT : 15

Objective : Single Inheritance

```
#include <iostream>
```

```
using namespace std;
```

```
class Person {
```

```
public:
```

```
    int id;
```

```
    char name[100];
```

```
    void set_pc() {
```

```
        cout << "Enter ID: ";
```

```
        cin >> id;
```

```
        cout << "Enter Name: ";
```

```
        cin >> name;
```

```
    }
```

```
    void display_pc() {
```

```
        cout << "\nID = " << id << endl;
```

```
        cout << "Name = " << name << endl;
```

```
    }
```

```
};
```

```
class Student : public Person {
```

```
public:
```

```
    int fee;
```

```
    char course[100];
```

```
    void set_sc() {
```

```

        cout << "Enter Fee: ";

        cin >> fee;

        cout << "Enter Course: ";

        cin >> course;

    }

    void display_sc() {

        cout << "Fee = " << fee << endl;

        cout << "Course = " << course << endl;

    }

};

int main() {

    Student obj;


    obj.set_pc();

    obj.set_sc();


    obj.display_pc();

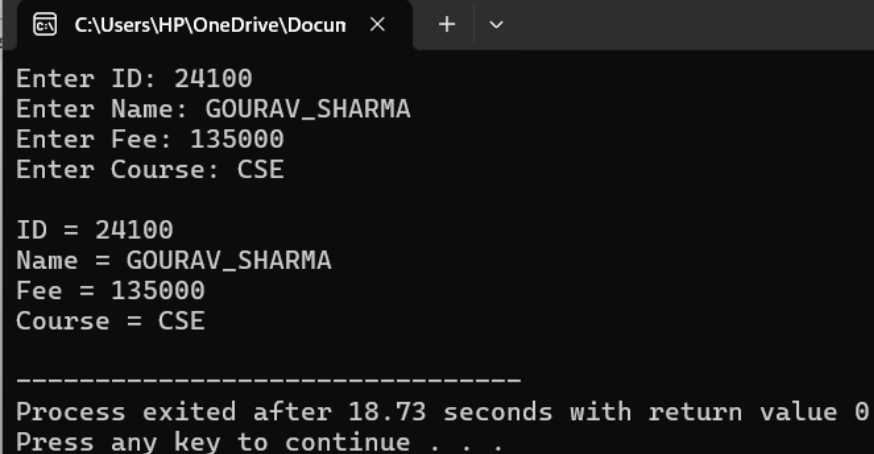
    obj.display_sc();


    return 0;

}

```

Output:



```

C:\Users\HP\OneDrive\Docun >
Enter ID: 24100
Enter Name: GOURAV_SHARMA
Enter Fee: 135000
Enter Course: CSE

ID = 24100
Name = GOURAV_SHARMA
Fee = 135000
Course = CSE

-----
Process exited after 18.73 seconds with return value 0
Press any key to continue . . .

```

EXPERIMENT : 16

Objective : Multiple Inheritance

Source Code:

```
#include <iostream>

using namespace std;

class Person {
public:
    int id;
    char name[100];

    void set_pc() {
        cout << "Enter ID: ";
        cin >> id;
        cout << "Enter Name: ";
        cin >> name;
    }

    void display_pc() {
        cout << "ID = " << id << endl;
        cout << "Name = " << name << endl;
    }
};

class College {
public:
    char college_name[100];

    void set_college() {
```

```

        cout << "Enter College Name: ";

        cin >> college_name;

    }

    void display_college() {
        cout << "College = " << college_name << endl;
    }
};

class Student : public Person, public College {
public:
    void show() {
        display_pc();
        display_college();
    }
};

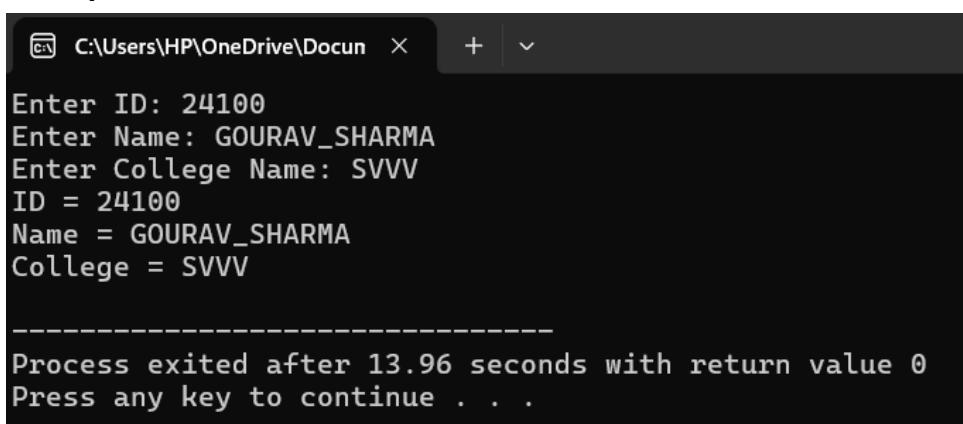
int main() {
    Student obj;

    obj.set_pc();
    obj.set_college();
    obj.show();

    return 0;
}

```

Output:



```

C:\Users\HP\OneDrive\Docun >
Enter ID: 24100
Enter Name: GOURAV_SHARMA
Enter College Name: SVVV
ID = 24100
Name = GOURAV_SHARMA
College = SVVV

-----
Process exited after 13.96 seconds with return value 0
Press any key to continue . . .

```

EXPERIMENT : 17

Objective: Hierarchical Inheritance

Source Code :

```
#include <iostream>

using namespace std;

class Person {
public:
    int id;
    char name[100];

    void set_person() {
        cout << "Enter ID: ";
        cin >> id;
        cout << "Enter Name: ";
        cin >> name;
    }

    void display_person() {
        cout << "ID = " << id << endl;
        cout << "Name = " << name << endl;
    }
};

class Student : public Person {
public:
    void display_student() {
        cout << "This is Student class.\n";
    }
}
```

```
};
```

```
class Teacher : public Person {
```

```
public:
```

```
    void display_teacher() {
```

```
        cout << "This is Teacher class.\n";
```

```
    }
```

```
};
```

```
int main() {
```

```
    Student s;
```

```
    Teacher t;
```

```
    cout << "Student Info:\n";
```

```
    s.set_person();
```

```
    s.display_person();
```

```
    s.display_student();
```

```
    cout << "\nTeacher Info:\n";
```

```
    t.set_person();
```

```
    t.display_person();
```

```
    t.display_teacher();
```

```
    return 0;
```

```
}
```

Output:

```
Student Info:
Enter ID: 24100
Enter Name: GOURAV_SHARMA
ID = 24100
Name = GOURAV_SHARMA
This is Student class.
```

```
Teacher Info:
Enter ID: 52100
Enter Name: DEVENDRA_SIR
ID = 52100
Name = DEVENDRA_SIR
This is Teacher class.
```

```
-----
Process exited after 48.68 seconds with return value 0
Press any key to continue . . . |
```

EXPERIMENT : 18

Objective : Hybrid Inheritance

Source Code :

```
#include <iostream>

using namespace std;

class Person {
public:
    int id;
    char name[100];

    void set_person() {
        cout << "Enter ID: ";
        cin >> id;
        cout << "Enter Name: ";
        cin >> name;
    }

    void display_person() {
        cout << "ID = " << id << endl;
        cout << "Name = " << name << endl;
    }
};

class College {
public:
    char cname[100];

    void set_college() {
```

```

        cout << "Enter College Name: ";
        cin >> cname;
    }

    void display_college() {
        cout << "College = " << cname << endl;
    }
};

class Student : public Person {
public:
    char course[100];

    void set_student() {
        cout << "Enter Course: ";
        cin >> course;
    }

    void display_student() {
        cout << "Course = " << course << endl;
    }
};

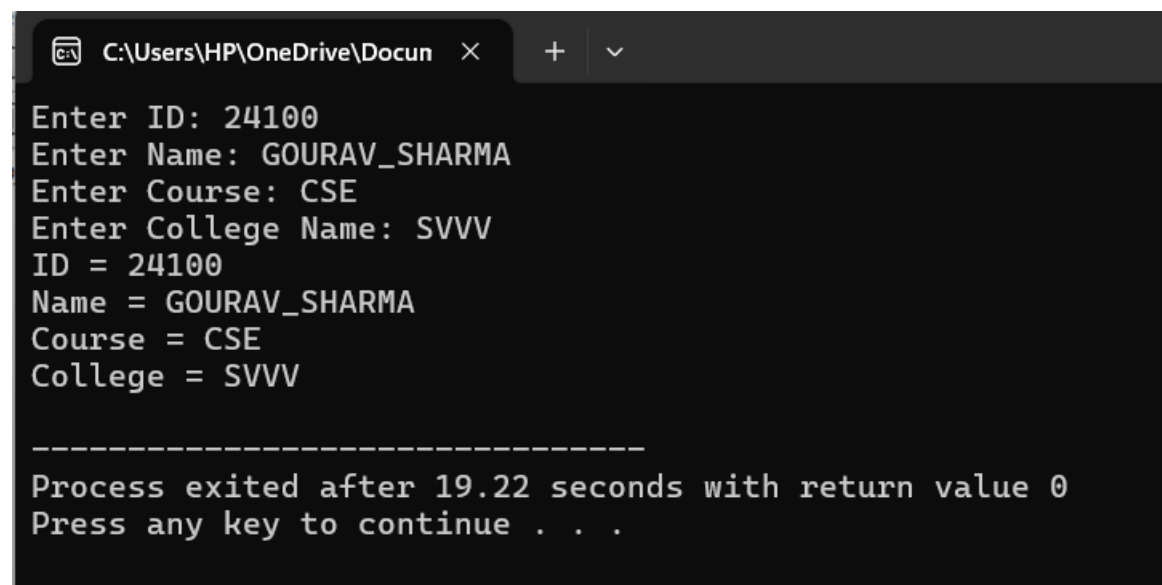
class Result : public Student, public College {
public:
    void show_all() {
        display_person();
        display_student();
        display_college();
    }
};

```



```
    }  
};  
  
int main() {  
    Result r;  
  
    r.set_person();  
    r.set_student();  
    r.set_college();  
  
    r.show_all();  
  
    return 0;  
}
```

Output:



```
C:\Users\HP\OneDrive\Docun  X  +  v  
Enter ID: 24100  
Enter Name: GOURAV_SHARMA  
Enter Course: CSE  
Enter College Name: SVVV  
ID = 24100  
Name = GOURAV_SHARMA  
Course = CSE  
College = SVVV  
  
-----  
Process exited after 19.22 seconds with return value 0  
Press any key to continue . . .
```